

Lippmann-Schwinger Equation

- QFT perturbation theory:

$H = H_0 + V$ where we know
Eig states of H_0

$$H_0 |q\rangle = E |q\rangle \quad \text{Eig state } H_0$$

$$\text{or } H |\psi\rangle = E |\psi\rangle \quad \text{Eig state } H$$

can write

$$|\psi\rangle = |q\rangle + \frac{1}{E - H_0} V |\psi\rangle$$

$\frac{1}{E - H_0}$ is a green func.

$\rightarrow$ L. S. kernel

$$\Pi_{LS} = \frac{1}{E - H_0}$$

- can use this for perturbation theory
(specifically QFT)

- scattering theory

- Inverse of $E - H_0$ is not well
defined in $\frac{1}{E - H_0 + i\epsilon}$

Cont:

define operator T as

$$V |\psi\rangle = T |q\rangle \quad \text{for}$$

$|\psi\rangle$ in terms $|q\rangle$

$$\rightarrow |\psi\rangle = |q\rangle + \frac{1}{E - H_0} T |q\rangle$$

$$V|\psi\rangle = V|\phi\rangle + V \frac{1}{E - H_0} V|\phi\rangle$$

w/ F $\rightarrow \langle \phi_j | V | \psi \rangle = \langle \phi_j | V | \phi \rangle + \langle \phi_j | V \frac{1}{E - H_0} V | \phi \rangle$

???

$$\rightarrow T = V + V \frac{1}{E - H_0} T$$

$$\rightarrow T = V + V \pi_{\perp} V + V \pi_{\perp} V \pi_{\perp} V \pi_{\perp} V + \dots$$

$\rightarrow$ invariant eigenvectors of $|\phi_j\rangle$

$$\langle \phi_j | \phi_j \rangle = \langle \phi_j |$$

$$\rightarrow \langle \phi_i | T | \phi_f \rangle = \langle \phi_i | V | \phi_f \rangle$$

$$+ \langle \phi_i | V \frac{1}{E - H_0} V | \phi_j \rangle \langle \phi_j | V | \phi_i \rangle + \dots$$

$$\text{or } T_{if} = V_{if} + V_{fi} \pi_{\perp} V_{ij} + V_{fi} \pi_{\perp} V_{jk} \pi_{\perp} V_{ki} + \dots$$

$$\text{w/ } \pi_{\perp}(k) = \frac{1}{E - E_k}$$

$$E_f = E_i$$

first term T_{ij} gives born approx,

Electron scattering off electron

Transition Matrix Element =

$$T_{fi} = V_{fi} + \sum_n V_{fn} \frac{1}{E_i - E_n} V_{ni} + \dots$$

Notes

- E_n energy of int. state ???

- E_i init energy

$|i\rangle = |\psi^1_e \psi^2_e\rangle$ and $\langle f| = \langle \psi^3_e \psi^4_e|$
both have 2 electrons.

- Coulomb law replaced by photon exchange
add $q(x)/photon$

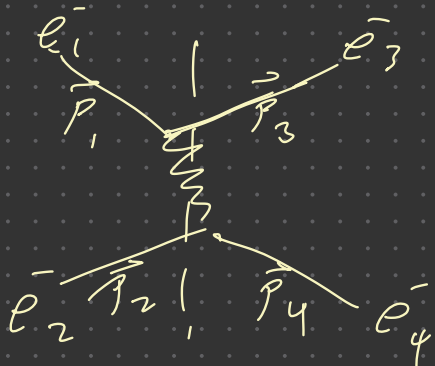
$$\Rightarrow V = \frac{1}{2} e \int d^3x \psi_e(x) q(x) \psi_e(x)$$

- $1/2$ comes from ignoring self

- Interaction turns $\vec{p}_1$ electron into
 $\vec{p}_3$ electron & $\vec{p}_2$ photon

- Both $|i\rangle$ & $\langle f|$ have 2 elec. & no phot
so $V_{ii} = 0$???

Kegman Diagram:



- In this state electrons interact via
Retarded Propagator

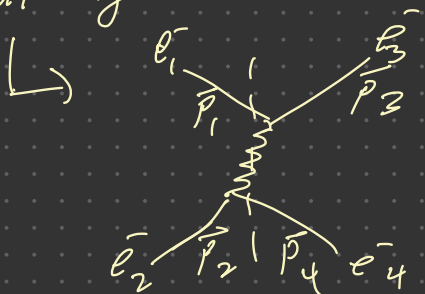
$$|n\rangle = |\psi_e^3 q^2 q_e^2\rangle, \quad q^2 \text{ phot. med.}$$

$$V_{ni}^2 = e \langle \psi_e^3 q^2 q_e^2 | V | \psi_e^1 q_e^2 \rangle$$

$$= e \langle \psi_e^3 q^2 | V | \psi_e^1 \rangle \langle q_e^2 | q_e^2 \rangle$$

$$= e \langle \psi_e^3 q^2 | V | \psi_e^1 \rangle$$

analogously



(Advanced)

$$w/V_{ni}^{(+)} = \langle \psi_e^4 q^2 | V | \psi_e^2 \rangle$$

$$V_{fi}^R = \langle \psi_e^R | V | \psi_e \rangle = e \int d^3r \langle \psi_e^R | \phi^2 | \psi_e(r) \phi(r) \psi_e(r) | \phi_e \rangle$$

Note:

earlier:

$$\phi(\vec{r}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{i\vec{p} \cdot \vec{r}} + a_p^\dagger e^{-i\vec{p} \cdot \vec{r}})$$

a.k.a.

$$\langle \phi^2 | \phi(\vec{r}) | 0 \rangle = e^{-i\vec{r} \cdot \vec{0}}$$

we obtain:

$$V_{fi}^{(R)} = e \int d^3r e^{i(\vec{p}_1 - \vec{p}_0 - \vec{p}_3) \cdot \vec{r}} \\ = e (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_3)$$

similar w/ other med for a decay
(such as $V_{fi}^{(A)}$, $V_{fi}^{(B)}$, etc.)

2??
"non resonant"

Have: $\text{Pr. of } \gamma \text{ decay}$

$$T_{fi} = \int d^3r \int d^3p (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) (2\pi)^3 \delta^3(\vec{p}_2 - \vec{p}_4 + \vec{p}_0) \frac{e^2}{E_i - E_f}$$

- Energy not conserved in intermediate states
since $E_u = E_i$ would mean explosion

Replacements:

Phot. energy $\rightarrow E_2$

Incoming electron energies $\rightarrow E_1, E_2$

outgoing electron energies $\rightarrow E_3, E_4$

Momentum conservation

$$\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4$$

photon momentum (by conservation):

$$\vec{p}_2 = \vec{p}_1 - \vec{p}_3$$

- photon is on shell $\Rightarrow p_2^2 = E_2^2 - \vec{p}_2^2 = 0$

$$E_2 = |\vec{p}_2| \quad \text{or} \quad p_2^2 = (E_2, \vec{p}_2) \cdot (E_2, \vec{p}_2) = (\vec{p}_1 - \vec{p}_3) \cdot (\vec{p}_1 - \vec{p}_3)$$

- Different int. state energies in retarded & advanced cases

Retarded:

ψ_e^3 already in E_3 so

$$E_u^{(R)} = E_3 + E_2 + E_2$$

Ignoring $(2\pi)^3$ factors,

$$\begin{aligned} T_{fi}^{(R)} &= \frac{e^2}{E_i - E_u^{(R)}} = \frac{e^2}{(E_1 + E_2) - (E_3 + E_2 + E_2)} \\ &= \frac{e^2}{(E_1 - E_3) - E_2} \end{aligned}$$

Advanced:

$$E_u^{(A)} = E_4 + E_1 + E_2$$

$$\begin{aligned} T_{fi}^{(A)} &= \frac{e^2}{E_i - E_u^{(A)}} = \frac{e^2}{(E_1 + E_2) - (E_4 + E_1 + E_2)} = \frac{e^2}{(E_2 - E_4) - E_2} \end{aligned}$$

Energy cons.

$$E_1 + E_2 = E_3 + E_4$$

$$\Rightarrow E_1 - E_3 = E_4 - E_2 = \Delta E$$

$$T_{fi}^{(R)} + T_{fi}^{(A)} = \frac{e^2}{E_i - E_2^R} + \frac{e^2}{E_i - E_2^A} = \frac{e^2}{\Delta E - E_2} + \frac{e^2}{-\Delta E - E_2}$$

$$= \frac{2e^2 E_2}{(\Delta E)^2 - (E_2)^2}$$

$$\text{pdf } K^\mu = p_3^\mu - p_1^\mu = (\Delta E, \vec{F})$$

$$\Rightarrow K^2 = (\Delta E)^2 - (E_2)^2$$

$$\rightarrow T_{fi} = T_{fi}^{(R)} + T_{fi}^{(A)} = 2E_2 \left(\frac{e^2}{K^2} \right)$$

Note:

$$T_{fi} \text{ has } \frac{1}{K^2} \stackrel{\text{L.T.}}{=} -\frac{1}{\square}$$

Feynman-ÖFT Rules

- All states are on shell
- Matrix elements $\langle i, j \rangle$ vanish ??? unless 3 momentum conserved
- Energy is NOT conserved per vertex

on-shell:

" satisfies free-field equations of motion

"Early Infrinites"

2nd quantized photon prob Hamiltonian

$$H = \int \frac{d^3k}{(2\pi)^3} \omega_k (a_k^\dagger a_k + \frac{1}{2})$$

$$E_0 = \langle 0 | H | 0 \rangle = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} |\vec{k}| = \infty$$

- vacuum energy not measurable
- measure differences

→ Renormalization

zero point energy has con seq. for Casimir effect.

QFT "Renormalization"

$$\Delta E_n = \langle \psi_n | H_{int} | \psi_n \rangle + \sum_{m \neq n} \frac{|\langle \psi_n | H_{int} | \psi_m \rangle|^2}{E_n - E_m}$$

- Have to sum over all $|\psi_m\rangle$
- finite # states under energy level E

$$E \rightarrow \infty, \frac{1}{E - E_m} \rightarrow 0$$

- ∞ states & phase space density $\sim \int d^3k \sim E^3$

$$\sim \frac{E^3}{E - E_m} \rightarrow \infty$$

??? at the gap frequency cutoff.

$$E_n \text{ Hint} = e \vec{E} \cdot \vec{r} = e |E| r$$

$$\hookrightarrow \langle \psi_n | \text{Hint} | \psi_n \rangle = 0$$

$$\rightarrow \Delta E_0 = \sum_{m \neq 0} \frac{|\langle \psi_0 | \text{Hint} | \psi_m \rangle|^2}{E_0 - E_m} = \underbrace{\sum_{m < 0}}_{\text{negative}} + \underbrace{\sum_{m > 0}}_{\text{positive}}$$

key: $\equiv$ are $\vec{E}$ sources

$\sum$ are photons

— between $\sum$ $\sum$ is

electron propagator
($\frac{1}{E_0 - E_m}$)

proach ΔE_0 finite:

pf: $E_0 < 0$ wlog, $E_m > E_1 > E_0$

$$\mathbb{1} = \sum_{m \neq 0} |\psi_m\rangle \langle \psi_m| = |\psi_0\rangle \langle \psi_0| + \sum_{m \neq 0} |\psi_m\rangle \langle \psi_m|$$

$$\begin{aligned} \text{??? How} \\ -\Delta E_0 &\leq \frac{1}{E_1 - E_0} \sum_{m \neq 0} \langle \psi_0 | \text{Hint} | \psi_m \rangle \langle \psi_m | \text{Hint} | \psi_0 \rangle \\ &= \frac{1}{E_1 - E_0} \left[\langle \psi_0 | \psi_{\text{Hint}}^e | \psi_0 \rangle - \langle \psi_0 | \text{Hint} | \psi_0 \rangle \right] \end{aligned}$$

PHS restriction #

with
2,2,2

$\therefore \Delta E_0$ bounded above & below (by 0)

E field created by e^-

— need photons created by e^-
annihilation of e^- & e^+ electrons

$$\rightarrow H_0 = H_{\text{atom}} + H_{\text{photon}}$$

2,2,2 (4.41)

$$H_0 |\psi_n, \{k_n\}\rangle = \left(E_n + \sum_k \omega_k a_k \right) |\psi_n, \{k_n\}\rangle$$

n_k excitations of photon, momentum $\vec{k}$

- number can only have (photon created & destroyed)
- integrate over momentum

Momentum cons.

$$\rightarrow \vec{p}_{\text{atom}} \sim - \vec{k}_{\text{photon}}$$

Excited energy 2,2,2

$$E \sim |\vec{p}| + |\vec{k}|$$

$$\rightarrow \Delta E_0 \sim \int \frac{d^3 \vec{p}}{(2\pi)^3} \int \frac{d^3 \vec{k}}{(2\pi)^3} \int d^3 k \frac{e^{i(\vec{k}-\vec{p})\cdot\vec{x}}}{E_0 - (|\vec{p}| + |\vec{k}|)}$$

2,2,2

$$\rightarrow \Delta E_0 \sim \int \frac{\sqrt{3}k}{(2\pi)^3} \frac{1}{|k|} = \frac{1}{2\pi^2} \int k dk = \infty$$

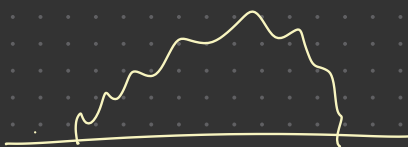
problems

- d^3k larger than $\frac{1}{|k|}$,
 - $\sum_{n \neq 0} |\psi_n\rangle \langle \psi_n|$ finite but sum over photon states diverges

- not fixed field, but dynamical fields w/ interaction state photons

$\langle \psi_0 | H_{int} | \psi_{int}; 1_n \rangle$ photons same as

$\langle \psi_{int} | H_{int} | \psi_0; 1_n \rangle$ photons $\rightarrow$ contractions

$$\Delta E_0 = \overline{\sum} \overline{\sum} \rightarrow$$


- opp publisher concluded QFT not to work

- missed other infinites, that caused this infinity
 e.g. electron mass